

Welcome to EECS 492 – Intro to AI!

EECS 492

Introduction to Artificial
Intelligence

Welcome to EECS. 492 – Intro to AI!

Download Today's
Slides & Set Up Poll
Everywhere (PV)

Slides available on Canvas

If Poll Everywhere attendance
doesn't work, come talk to me after
class (so I can mark down your
attendance)

Today's Class

- Meet the instructors
- What is AI?
- Course policies, syllabus, etc.
- A little bit of history

Meet the Staff

Elizabeth (Liz) Bondi-Kelly

- Assistant Professor in CSE
- PhD from Harvard University, advised by Prof. Milind Tambe
- Hometown: Dansville, NY
- Mom of 2!

<https://sites.google.com/view/elizabethbondi>

ecbk@umich.edu

| AI for Social Impact

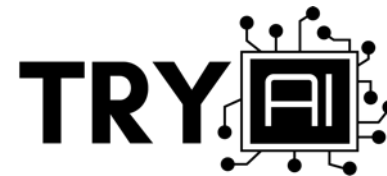
Public Health



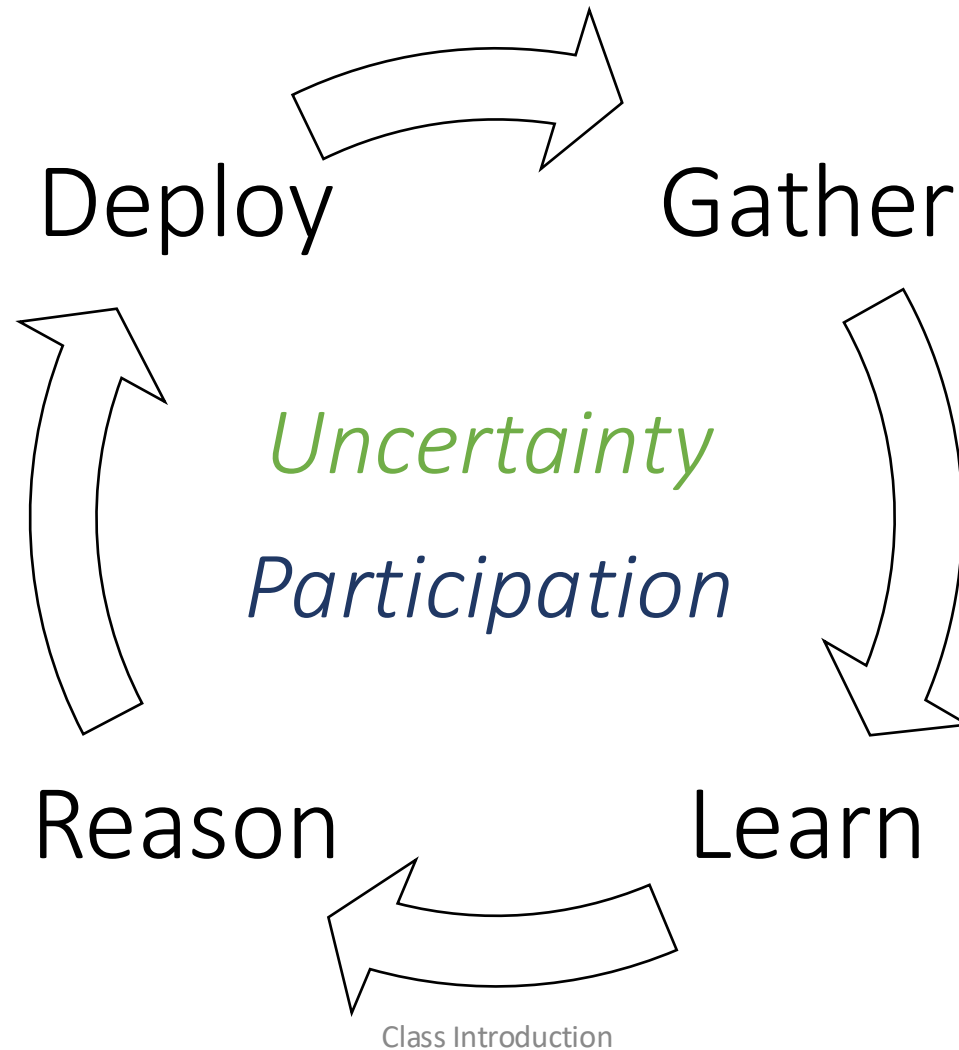
Conservation



Education



| AI for Social Impact Cycle



Research Question

How to account for **uncertainty** and **human participation** *throughout* the AI for social impact cycle?

PATHFinder Care Conversations testuser

Prenatal Care Questions Care Plan Hide Care Plan New Session

Report 100% Export PDF

Your Prenatal Care Plan

Prepared for test test - August 26, 2026

This plan is a starting point based on what you shared today. It is normal for it to change as your pregnancy progresses. You and your care team can review and adjust it together at your visits.

Key Dates

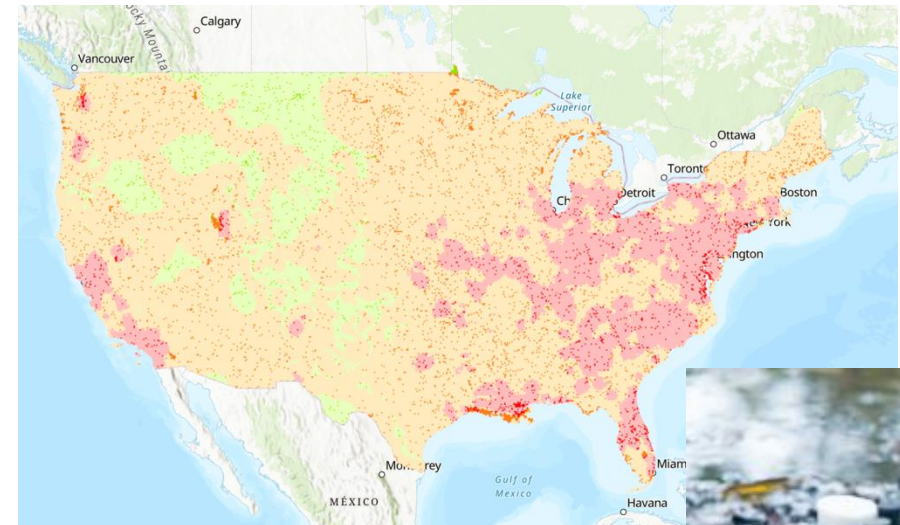
Date	Milestone
August 26, 2026	Today; estimated 8 weeks pregnant based on LMP
August 26–September 8, 2026	Intake and first prenatal visit window
September 9, 2026 onward	Cell-free DNA screening may be available if desired
September 16–October 6, 2026	Nuchal translucency ultrasound window if chosen

Ask About Report

Type a question. Drag-select text in the report for context. Double-select starts a new conversation.

Select text from the report and ask follow-up questions.

Ask about your report or timeline...



Instructor: Alexander Rodríguez

- Assistant Professor in CSE
- PhD in CS at Georgia Tech
- Hometown: Lima, Peru
- Hobbies: reading, hiking, dancing



Research Group: Complex AI

Research focus: ML and data mining

- **Time series:** forecasting, causal inference
- **AI for science:** hybrid models, differentiable simulators, AI agents
- **Uncertainty quantification:** conformal prediction
- **Multiagent systems:** agent-based modeling

Applications: Health and engineering

alrodri.engin.umich.edu



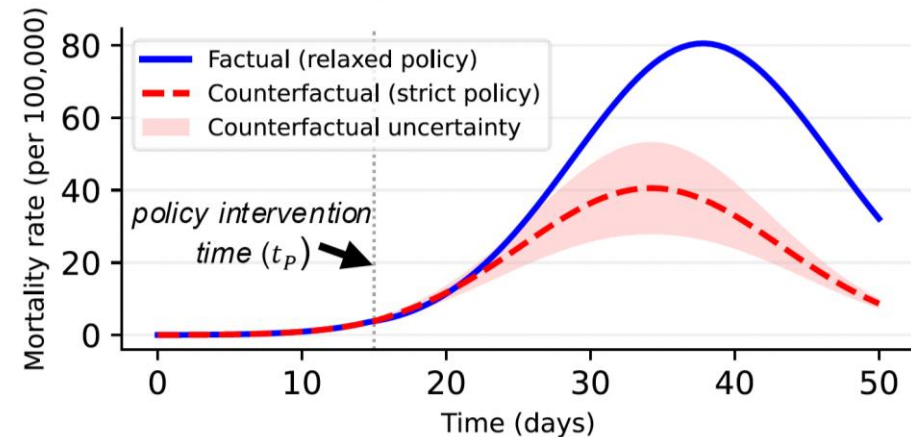
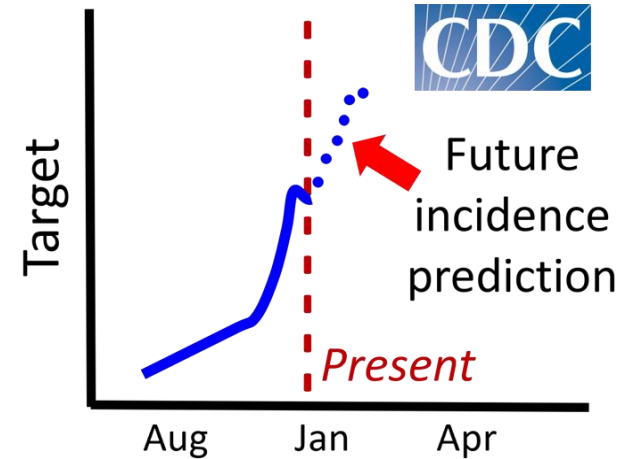
@_arodriguezca



alrodri@umich.edu

Time Series Predictive Modeling for Health

- Forecasts:
 - **Most likely outcome** based on past data (and/or beliefs)
- Projections:
 - Outcomes for **specific scenario** (e.g., treatment)
 - Require causal/mechanistic understanding



CSE 550: AI for Science – Winter 2027

Instructor: Alexander Rodríguez, Computer Science

How can AI enhance – not replace – scientific principles? This course explores recent advances that *integrate AI with simulations, equations, causal relations, and invariances* to solve real-world scientific problems.



<http://bit.ly/4kxoz46>



Open to graduate students in CSE, engineering, and sciences.

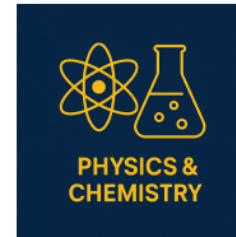
Core Methods: We examine methodological advances and their impact across scientific domains, including:

- Physics-informed neural networks
- Neural operators
- Differentiable simulators
- Surrogate modeling for simulation
- Knowledge-guided ML
- Discrepancy modeling
- Learning scientific equations
- Foundation and generative models

Cross-Cutting Topics:

- Uncertainty quantification
- AI safety and explainability

Scientific Applications: The course is domain-agnostic but draws examples from various fields, including:



Enroll this Winter and explore the frontiers of AI in science!

EXPLORE RESEARCH. EXPLORE MENTORSHIP.
EXPLORE YOU.

Explore Computer Science Research (2026–2027)

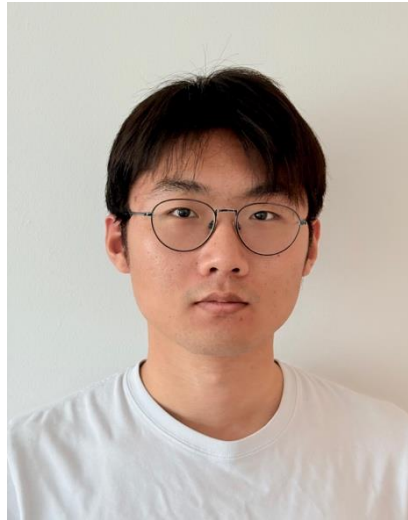
You'll be paired with a research mentor, contribute 6-10 hours per week to your project, and take part in events, seminars, and more!

The program welcomes undergraduate students from all years and backgrounds. We especially encourage women and students from groups underrepresented in computing to apply.

HOW TO APPLY? Go to ecsr.engin.umich.edu



Course Staff



Ruipu Li (GSI)



**Vaibhav Balloli
(GSI)**

What is AI?

Multiple Views on Intelligence

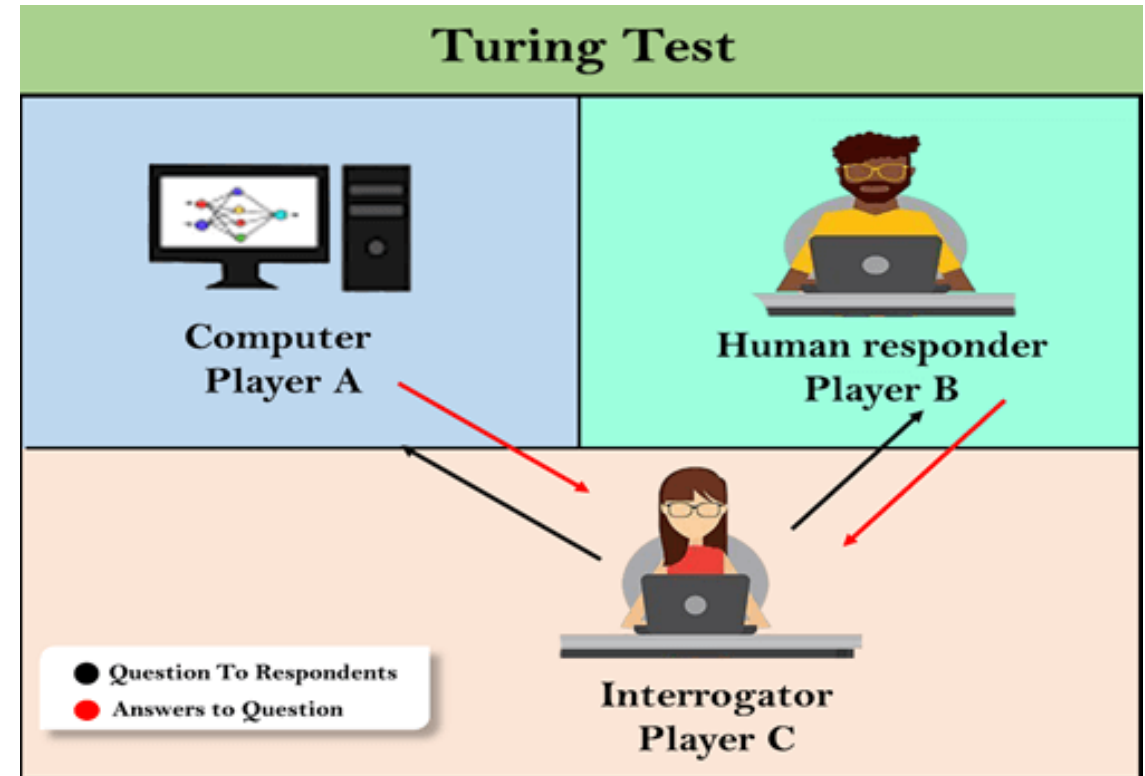
What is Artificial Intelligence?



Acting Humanly

- How can we define intelligence?
 - Turing Test
- Skills:
 - Natural language processing
 - Knowledge representation
 - Automated reasoning
 - Machine learning

Thinking Humanly	Thinking Rationally
Acting Humanly	Acting Rationally



<https://www.javatpoint.com/turing-test-in-ai>

Acting Humanly

Thinking Humanly	Thinking Rationally
Acting Humanly	Acting Rationally

- Total Turing test = includes interacting with objects & people
 - Computer vision
 - Speech recognition
 - Robotics
- What are some issues with this test?

Acting Humanly

Thinking Humanly	Thinking Rationally
Acting Humanly	Acting Rationally

The quest for “**artificial flight**” succeeded when the engineers and inventors **stopped imitating birds** and started using wind tunnels and learning about aerodynamics. Aeronautical engineering texts do not define the goal of their field as making “machines that fly so exactly like pigeons that they can fool even other pigeons.”

- Russell & Norvig, pg. 2 (our textbook)

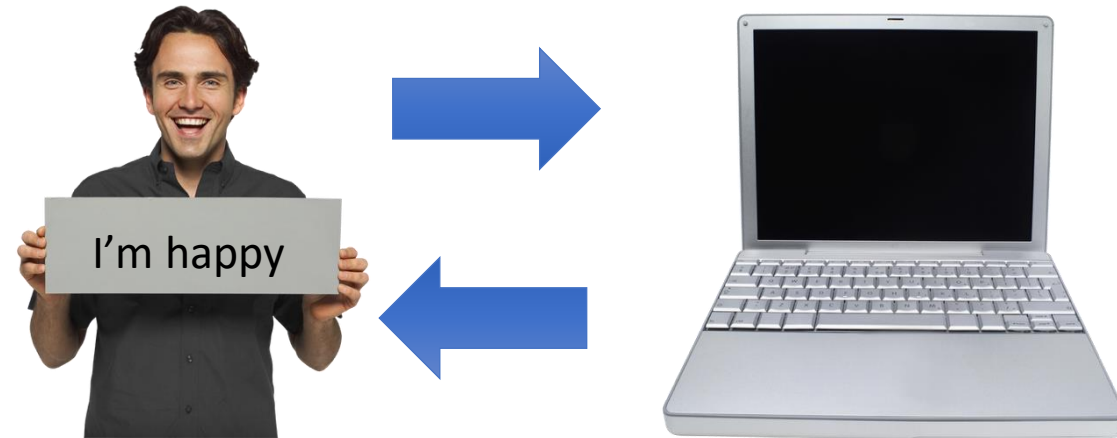
What is Artificial Intelligence?



Thinking Humanly

Thinking Humanly	Thinking Rationally
Acting Humanly	Acting Rationally

- First, we must know how humans think
 - Introspection
 - Psychological experiments
 - Brain imaging
- What can we learn about humans?
- **Cognitive science** – interdisciplinary field brings together computer AI models & experimental techniques from psychology



What is Artificial Intelligence?



Thinking Rationally

Thinking Humanly	Thinking Rationally
Acting Humanly	Acting Rationally

- Logic – provides precise notation for statements
- Example:
 - *Socrates is a man*
 - *All men are mortal*
 - Then, *Socrates is mortal*
- What are some limitations?

What is Artificial Intelligence?



Acting Rationally

- Agent – something that acts
 - What is a **rational agent**?
 - What is the goal of a rational agent?

Thinking Humanly	Thinking Rationally
Acting Humanly	Acting Rationally

Acting Rationally

- Agent – something that

Prevailing Approach
throughout Much of AI's
History

Thinking Rationally

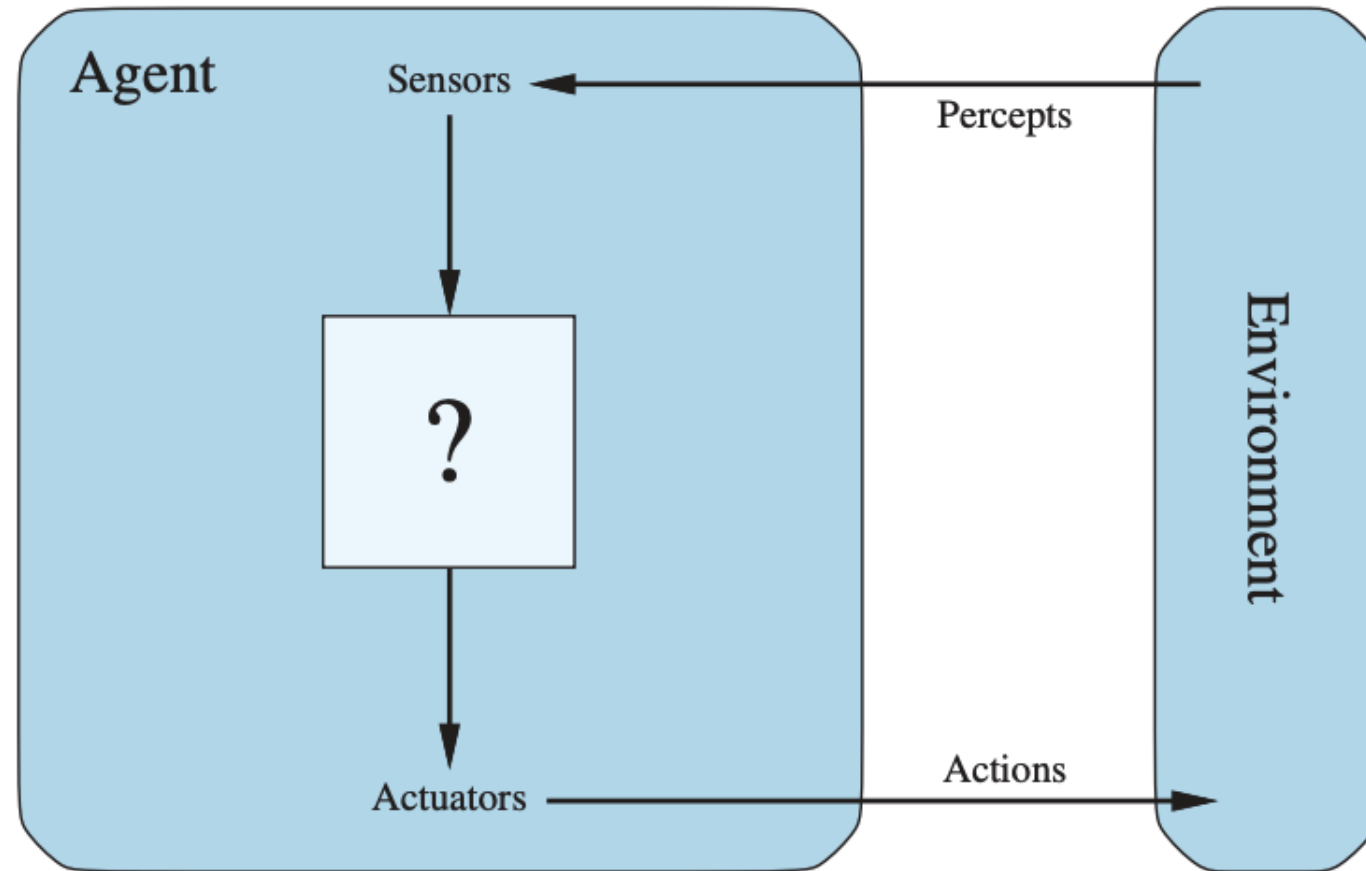
Acting Rationally

Acting Rationally

Thinking Humanly	Thinking Rationally
Acting Humanly	Acting Rationally

- Agent – something that acts
 - What is a **rational agent**?
 - What is the goal of a rational agent?
- Rational agent approach – make the best decision given uncertainty

This Course: Designing Rational Agents



What happens if we...

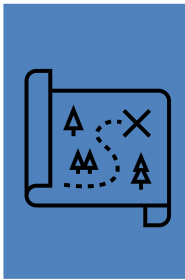
- Are time constrained?
- Choose an objective for the system?

From Rational Agents to Beneficial Machines

- What's wrong with the prevailing “acting rational” approach?
 - Assumes that we can specify our objective completely
 - What is the objective for a self-driving car?
- **Value alignment problem** = values or objectives put into the machine must be aligned with those of the human.
- --> The machine is uncertain about what our objectives are
 - Machine should have incentives to act cautiously, ask permission, defer to human control
- Should be thinking intentionally about *what* systems are doing

Course Policies, Syllabus, etc.

Big Themes of this Course



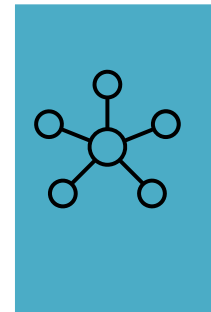
Search



Problem Solving



Uncertainty



Learning

What is EECS 492?

- A first course in Artificial Intelligence, providing experience in a range of key ideas.
- Objectives – finish the course with:
 - Specific modeling and analytical skills (e.g., search, logic, probability, ...)
 - Knowledge of many of the most important knowledge representation, reasoning, and machine learning schemes
 - General understanding of AI principles and practice
- Prepare for:
 - Further work in AI
 - Any work involving the design of computer programs for substantial applications

EECS 492 Topics

Boundaries are fuzzy!

A.I.

- Search
 - Uninformed
 - Informed
 - Local
 - Adversarial
- Constraint Satisfaction
- Logic
 - Propositional
 - First-Order
- Game theory

Machine Learning

- Decision Trees
- Linear Regression
- k-Nearest Neighbor
- SVMs
- Markov Decision Processes
- MCTS
- Reinforcement Learning

Neural Networks

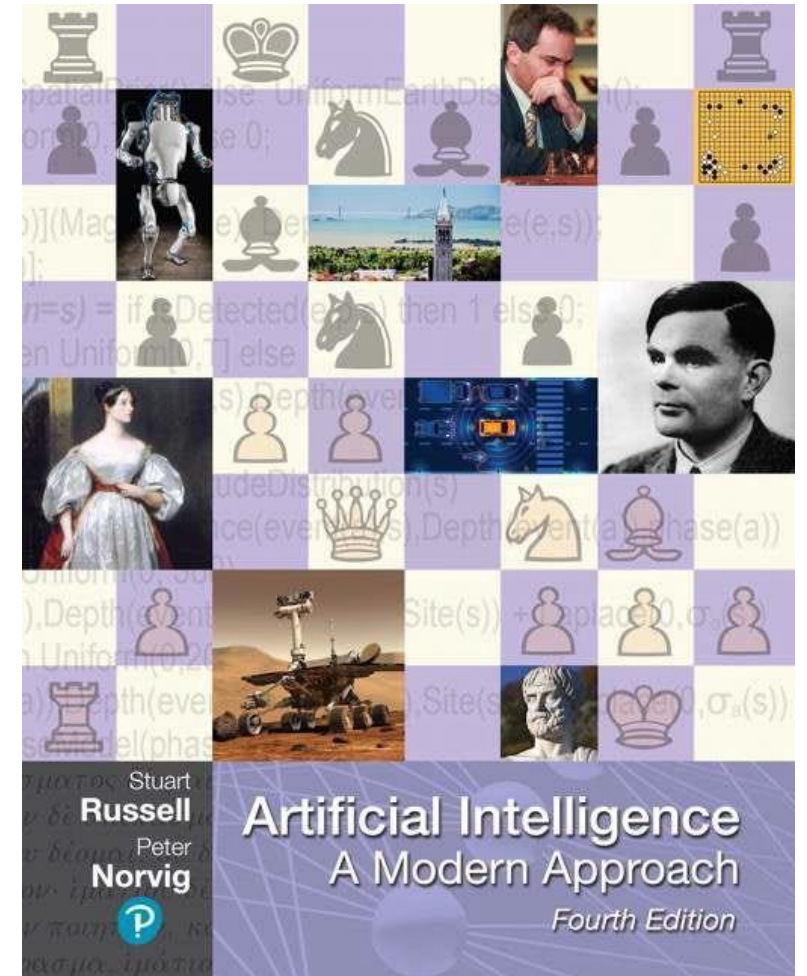
- Feed forward (& deep)
- Convolutional
- Recurrent
- Transformers
- Generative

Topics subject to modest change

Lots of math across the entire course!

The Textbook

- Russell and Norvig:
Artificial Intelligence: A Modern Approach
(fourth edition)
- Encouraged, not required



Lectures

- Attendance will be taken using Poll Everywhere
 - Required in-person attendance: 14 out of the 27 lectures
- Built in flexibility – no exceptions will be granted except in extenuating circumstances
- Lectures will be recorded

Discussion Sections

- We want discussion sections to be a valuable part of this course for you!
 - Current events in AI; real-world problems and solutions
 - New exercises to practice the material
 - Review of old material (e.g., basic search algorithms, probability) that we will use in 492
 - Discussion section content won't be a repeat of lecture content
- Ahead of time, we will tell you what discussion section will cover, so you know if it will be valuable for you to attend

Discussion Sections

- Attendance not required
- Discussion sections will **not** be recorded; slides will be available
- Try to join a consistent discussion section to get to know classmates and your instructor

Friday 10:30am- 11:30am	1003 EECS	Vaibhav Balloli
Friday 1:30- 2:30pm	1005 EECS	Vaibhav Balloli

Office Hours (starting next week)

Monday:

Liz Bondi-Kelly (conceptual questions only), 12:00-12:30 PM, 3160 Leinweber
Ruipu Li, 4:30-5:00PM, 1637 BBB Table #1

Tuesday:

Alexander Rodríguez (conceptual questions only), 3:30-4:00 PM, 3757 BBB
Vaibhav Balloli, 4:00-4:30PM, 3270 Leinweber

Wednesday:

Liz Bondi-Kelly (conceptual questions only), 12:00-12:30 PM, 3160 Leinweber

Thursday:

Alexander Rodríguez (conceptual questions only), 3:30-4:00 PM, 3757 BBB

*More office hours can be arranged by appointment

- Available in syllabus and course calendar; changes will be notified

Staying Connected

- For **lecture content** questions that arise outside of class:
 - Office hours
 - and/or
 - [Piazza](#) ➡
- For questions about how to solve a **homework** problem:
 - Office hours
 - and/or
 - [Piazza](#) ➡ if you can ask without sharing your answer and without sharing code
- For questions about course **policies**:
 - [Piazza](#) ➡ if it's not personal
 - or
 - [Admin form](#) ➡ if it's personal.
- For concerns about how something was **graded**:
 - See "Grading" section in the syllabus.
- For **personal** questions regarding:
 - Joining the course late
 - Sickness or another life challenge interfering with your work
 - SSD accommodations
 - Scheduling problems
 - Any other non-content-related personal question or concern
 - → [Admin form](#) ➡



piazza

Staying Connected

- Canvas:
 - Slides
 - Homework assignments
- Piazza:
 - Main discussion forum, (much) faster than email!
 - Please post questions/ideas/thoughts there
 - You can post anonymously too (to your classmates)



Assignments

- Homeworks – Three throughout the semester
 - Conceptual questions and coding
- Must **work alone** for conceptual questions (written part)
 - OK to discuss ideas and high-level concepts w/ classmates
- Can **work with a partner** for coding



Assignments

- Ethics assignments – 2 throughout the semester
 - Required to read paper; you'll do a quiz before the discussion
 - Required to attend your discussion section
 - The section you enrolled to avoid overflow in the classroom
- If you can't attend a discussion, must come to faculty office hours to talk about the reading within a week

Late Policy

- Late policy (for all assignments)
 - Minus 10% per day, up to 3 days
- Every student gets 5 late-day tokens to use for any HWs
 - No questions asked!
 - Hours are rounded up. So if you submit one HW 10hrs or 1hr late, it will be counted as 1 late day
- Medical or other emergencies are always excused separately: please fill out our admin form, always with documentation

Generative AI Policy

- Can use generative AI for conceptual questions
 - Need to include an explanation (see syllabus)
- CANNOT use generative AI for programming

Exams

- Both will be in person – make travel plans accordingly
- **Midterm exams:** Midterm #1 will take place on the week of October 14, 6:00 - 8:00pm, and Midterm #2 will take place the week of November 18, 6:00 - 8:00pm.
- **Final exam:** The final exam will take place on Thursday, December 17, 10:30 am- 12:30 pm. The final exam will be cumulative (it will cover all topics in the course).

Wellness

- If for any reason you are having difficulty in this course, please come talk to me; I want to help.

Grading

- Homeworks 30% (12.5% each for 4 assignments)
- Ethics assignments 10% (5% each for 2 assignments)
- Midterm #1 15%
- Midterm #2 15%
- Final 25%
- Attendance 5%
- Course evaluations +1% (extra credit, 0.5% for each evaluation – midterm and final)

Some advice on how to succeed in EECS 492

- Come to lectures regularly
 - Get to know the people around you so that you have friends to complete the homeworks with and study with
- Come to discussion sections regularly
 - Smaller setting to ask lots of questions
 - Get to know your GSI / IA
- If you're worried about falling behind, come to office hours regularly, even if you don't have specific questions
 - Use office hours as a time to work on your homework alongside other students

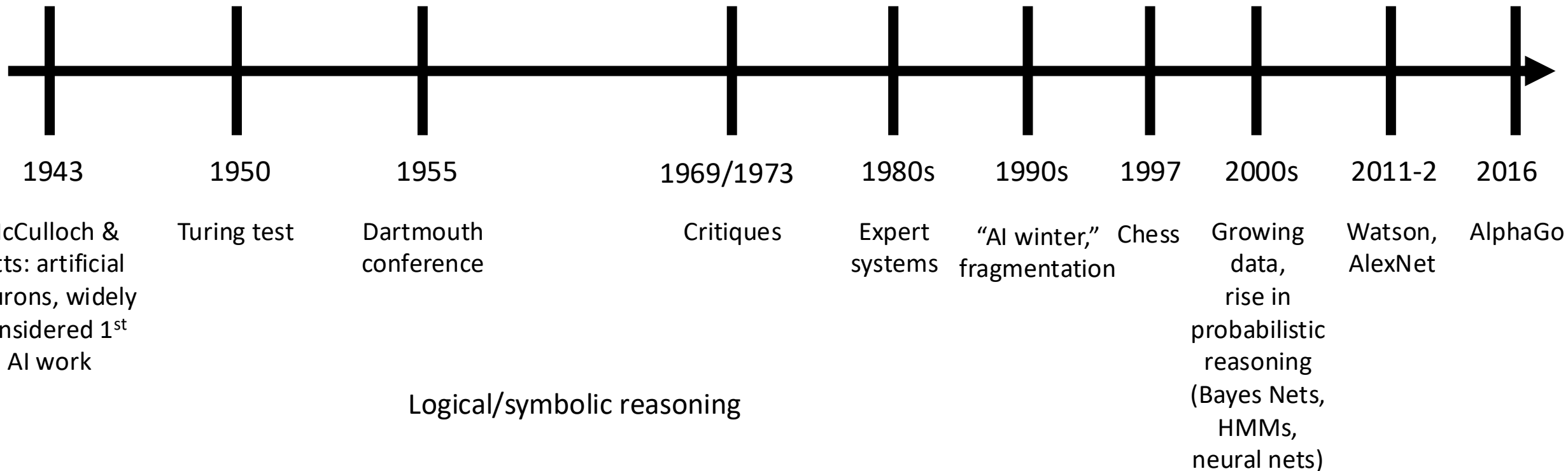
Some advice on how to succeed in EECS 492

- If something is causing you to fall behind in the class (a personal situation, an academic situation, etc.), come talk to me sooner rather than later!
 - I want you to succeed in this class, and I can help you make a plan to get back on track
- Start the assignments early – debugging takes time
- Start studying for the exams early

A Little Bit of History

A brief history of AI

Enthusiasm!



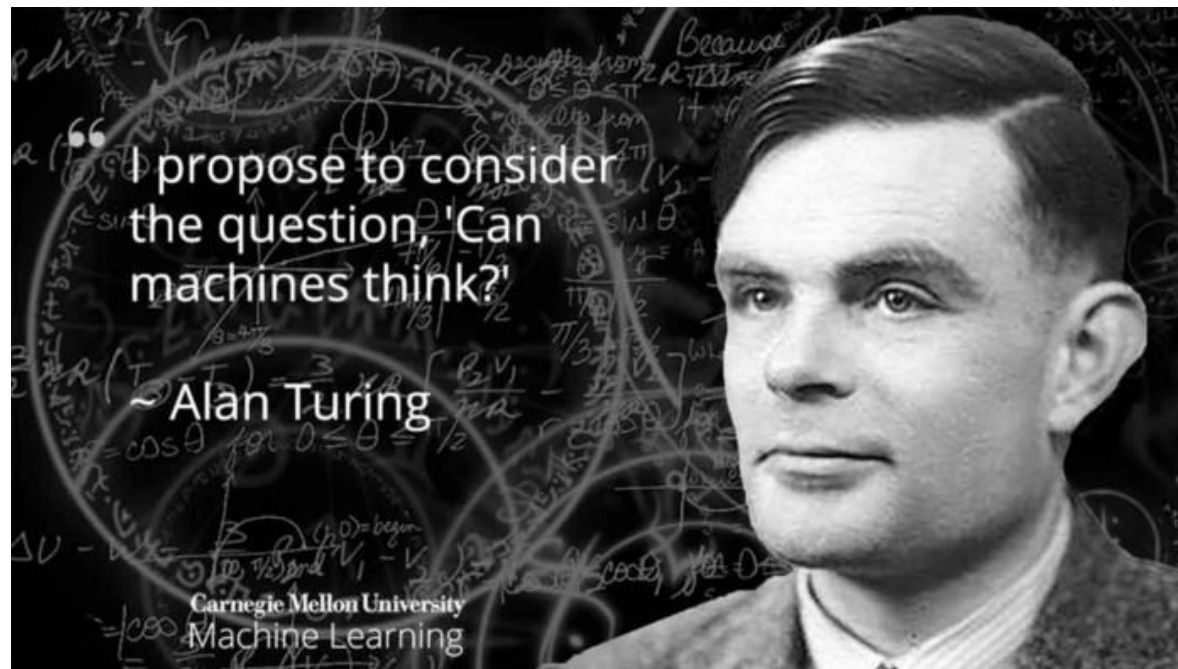
The beginning of AI...

- The neural network is born
 - 1943: Inspired by the brain, Warren McCulloch and Walter Pitts publish first mathematical model of a neural network
 - 1950: Marvin Minsky and Dean Edmonds build the first neural computer, the SNARC
 - 3,000 vacuum tubes
 - A surplus automatic pilot mechanism from a B-24 bomber
 - Network of only 40 neurons



The beginning of AI...

- Alan Turing's vision for AI
 - 1950 article – “Computing Machinery and Intelligence”
 - Introduced the Turing test, machine learning, genetic algorithms, reinforcement learning



The beginning of AI....

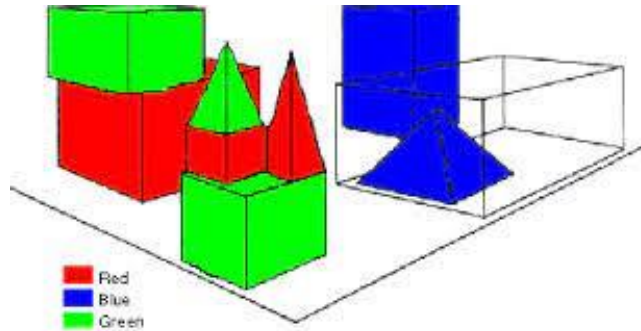
- Name “artificial intelligence” coined in 1956 Dartmouth conference:

“... The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer.”

They didn't quite finish...

“Look, Ma, no hands!”

- At the beginning, there was success was small, simple problems
 - 1952: Checkers-playing program performed at strong amateur level using early reinforcement learning (Arthur Samuel)
 - 1960s: Marvin Minsky worked on limited domains called **microworlds** (most famous: blocks world)



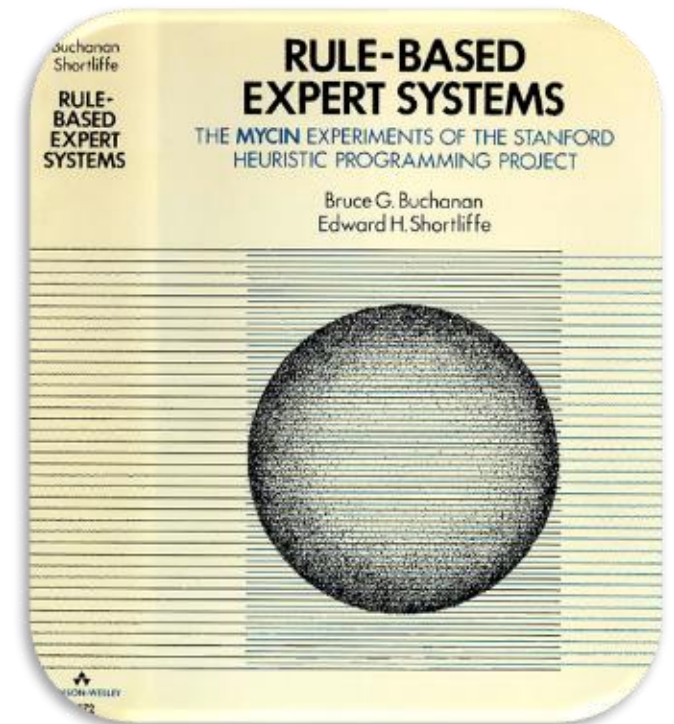
- 1958: John McCarthy creates Lisp programming language - dominant AI programming language for 30 years

A Dose of Reality (1966-1973)

- Early programs had no subject knowledge
 - Success via simple syntactic manipulations
 - “The vodka is good but the meat is rotten.”
- Many problems were harder than they looked
 - Being able to solve something in principle (on a small example) doesn’t mean that you can solve it practically in the real world

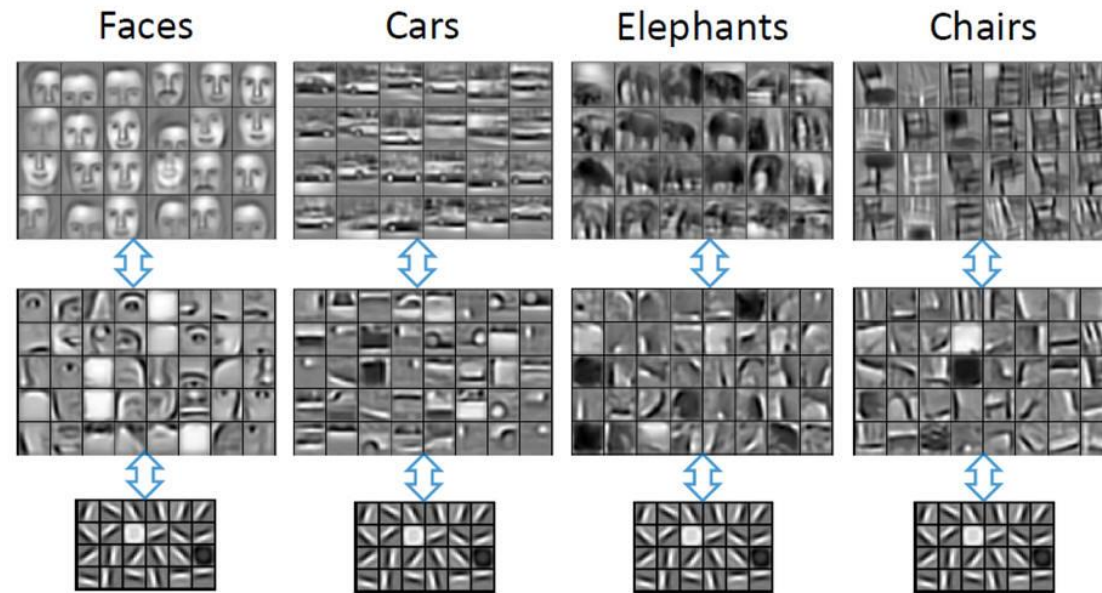
Expert Systems (1969-1986)

- Moving from general-purpose systems to **knowledge-intensive** systems in very specific domains
- Example: Mycin
 - 450 rules to diagnose blood infections
 - Issue – no general framework
 - Required expert knowledge from humans
 - Must capture uncertainty: introduced certainty factors
- Led to AI Winter: Companies failed to deliver on extravagant promises



The Return of Neural Networks (1986–now)

- Back-propagation was rediscovered in the mid-1980s!
 - Originally discovered in the early 1960s, but forgotten about
 - Enabled training of complex neural networks!



<http://youqianhaozhe.com/research.htm>

Probabilistic Reasoning and Machine Learning (1987-now)

- AI systems began to be grounded in more mathematical theory

- Probability rather than Boolean logic
- Machine learning rather than hand-coding
- Experimental results rather than philosophical claims



- Large benchmark problem sets established

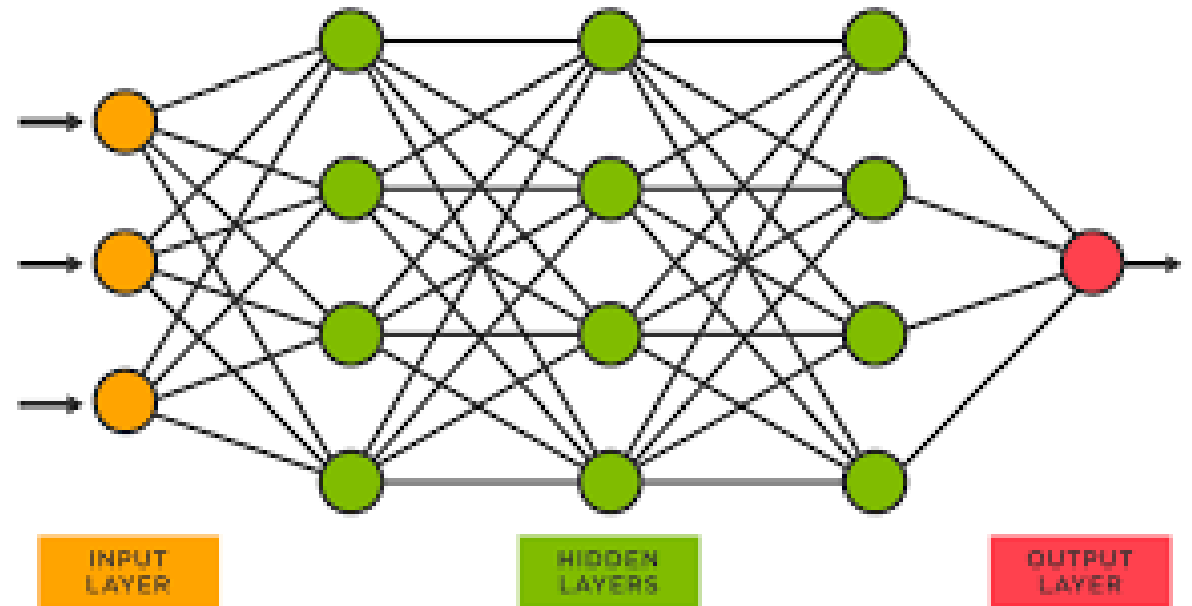
- AI began to draw more from other academic fields (e.g., statistics, decision theory)

Big Data (2001-now)

- Huge data sets are now available, thanks to the Internet and advances in computing power
 - Trillions of words of text
 - Billions of images
 - Billions of hours of speech video
 - So Much Data: genomic data, vehicle tracking data, clickstream data, ...
- Banko and Brillo (2001) – improvement in performance from increasing the amount of data is 2 or 3 orders of magnitude more than improving the algorithm (!!!)

Deep Learning (2011-now)

- **Deep learning** – machine learning using multiple layers of simple, adjustable computing elements (neurons)
- Deep learning needs:
 - Powerful hardware
 - Large amounts of training data



<https://www.tibco.com/>

Where is AI Now?

- Human performance or better in chess, Go, poker, Pac-Man, Jeopardy!, skin cancer detection, prostate cancer detection



Where is AI Now?

- Self-driving cars: Waymo vehicles have driven 10 million miles on public roads without a serious accident (human stepping in every 6,000 miles)



Final Reminders

- Office hours start next week (Tuesday)
- Read the syllabus
- ★ We are looking forward to having a wonderful semester learning with all of you 😊